

HEALTHCARE ANALYTIS

TEAM G-10

Diabetes Readmission Analysis

*Analyzing Clinical & Demographic Patterns
to Reduce 30-Day Hospital Readmissions*

Team Members

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GitHub: [SectionCG10_DiabetesReadmission](#)

Tableau Dashboard



Context & Problem Statement

Why this matters — and what we set out to answer

Sector Context

- US hospitals face multi-million dollar penalties under the Hospital Readmissions Reduction Program (HRRP)
- Avoidable 30-day readmissions signal failures in discharge quality and post-acute care
- Diabetic patients face compounded risk due to multiple comorbidities (circulatory, metabolic)
- Value-based care mandates demand measurable outcome improvements

Problem Statement

- How can clinical history, medication patterns, and demographics accurately predict 30-day readmission risk in diabetic patients?
- Which patient segments are most likely to return within 30 days of discharge?
- What specific clinical interventions reduce return rates most efficiently?

Objective

- Decision: Move from generic discharge protocols to a risk-stratified approach
- Support CMOs in identifying high-risk patients before discharge
- Empower CFOs with data-backed penalty avoidance strategies
- Provide Discharge Planners with actionable, patient-level risk flags

Data Engineering: Source to Pipeline

UCI Diabetes 130-US Hospitals Dataset | Python ETL | GitHub

101,763

Patient Encounters

54

Columns (Post-Clean)

10 Yrs

Data Span (1999–2008)

130

US Hospitals

ETL Pipeline Steps:

1

Extract

Ingested raw UCI dataset (101,766 rows × 50 cols). Notebook: 01_extraction.ipynb

2

Clean

Converted '?' → NaN. Dropped Weight (>90% null) & Medical Specialty (>40% null). Removed 3 duplicate rows.

3

Standardize

Mid-point mapped age bins ([70-80) → 75). Mapped Admission Type IDs to readable strings (e.g., '1' → 'Emergency').

4

Feature Engineer

Classified 700+ ICD-9 codes into 17 clinical groups (Circulatory, Respiratory, Metabolic...) via Python dict logic. Capped meds at 99th percentile (43).

Python: ICD-9 Diagnosis Mapping (02_cleaning.ipynb)

```
def classify_icd9(code):  
    val = float(code)  
    if 390 <= val <= 459 or val == 785: return 'Circulatory'  
    elif 460 <= val <= 519 or val == 786: return 'Respiratory'  
    else: return 'Other'
```

Key Data Dictionary Fields

- readmitted_under_30 — Binary target (1 = readmitted <30 days)
- prior_inpatient — Prior admissions in the past year
- diag_1_group — Mapped ICD-9 clinical category
- num_medications — Count of distinct drugs prescribed

KPI & Metrics Framework

Primary, Secondary, and Tertiary KPIs — linked to the business problem

101,763

Patient Encounters

71,515

Unique Patients

11.16%

30-Day Readmission Rate

Primary KPI: 30-Day Readmission Rate (%)

- Link to Problem: Directly measures discharge failure and HRRP penalty risk
- Formula: $\text{SUM}(\text{readmitted_under_30}) / \text{COUNT}(\text{encounters})$
- Computation: Python Pandas aggregations — notebook 03_eda.ipynb
- Benchmark: Current rate 11.16%; target is <9% post-intervention

Secondary KPI: Average Length of Stay (LOS)

- Link to Problem: Shorter stays increase efficiency but may raise readmission risk if premature
- Formula: $\text{AVG}(\text{time_in_hospital})$ across all encounters
- Computation: Aggregated from encounter records; segmented by diagnosis group
- Finding: Readmitted patients averaged 0.8 days longer than non-readmitted

Tertiary KPI: Medication Change Rate (%)

- Link to Problem: Acts as a proxy for clinical instability during the hospital encounter
- Formula: $\text{SUM}(\text{change} == \text{'Ch'}) / \text{COUNT}(\text{encounters})$
- Computation: Derived from medication change field in 03_eda.ipynb
- Finding: Change group had 11.82% readmission vs. 10.59% for no-change group

Key Insights: Exploratory Data Analysis

Translated to business language — from 03_eda.ipynb

High-Frequency Returners

01

Patients with ≥ 3 prior inpatient visits have a ~50% readmission rate — 4.5× higher than the general cohort average of 11.16%. This single segment drives disproportionate penalty exposure.

Circulatory Conditions Dominate

02

Circulatory diagnoses (heart failure, arrhythmia) account for ~30% of all 30-day returns. The 75–85 age group with circulatory comorbidities is the highest-volume risk cluster.

The Young-Adult Risk Paradox

03

A secondary risk peak exists at age 25–35, likely driven by unmanaged Type 1 Diabetes and a failure in outpatient support systems for young adults post-discharge.

Mid-Length Stay Is the Danger Zone

04

Most readmissions occur among patients who stayed 4–7 days — not extreme outliers. This 'mid-length' window is the critical discharge planning moment.

Medication Complexity = Risk

05

Strong positive correlation between number of medications and readmission status. More drugs signal more complex patients — 20+ medications increases risk materially.

Emergency Admissions Lead Risk

06

53.05% of all encounters are Emergency admissions. ER-based readmissions are more likely to involve Circulatory and Metabolic diagnoses than elective admissions.

Advanced Analysis: Statistical Validation

Hypothesis Testing · Feature Importance · Segmentation — scipy, statsmodels, sklearn

Mann-Whitney U Test

$P < 0.0001$

Tested whether prior inpatient visit count differs significantly between readmitted vs. non-readmitted groups. Result confirms prior history is the single strongest clinical predictor of return visits.

Independent T-Test (Length of Stay)

$\Delta = +0.8$ days

Readmitted patients had an average LOS 0.8 days longer than non-readmitted. Insight: Longer stays reflect higher underlying patient severity — they are NOT preventing readmissions.

Chi-Square Test (Admission Type)

$P < 0.01$

Confirmed statistically significant association between admission type (Emergency vs. Elective) and 30-day readmission status. Emergency admissions carry materially higher return risk.

Random Forest Feature Importance (sklearn)



New Understanding from Advanced Analysis

- Prior visits is not just correlated — it is 4.5× more predictive than any demographic variable
- Longer LOS does NOT reduce readmissions — it reflects severity, not quality
- Emergency admission + Circulatory diagnosis = highest-risk compound segment
- Libraries: scipy.stats (U-test, t-test), sklearn (Random Forest), pandas (aggregations)

Tableau Dashboard Walkthrough

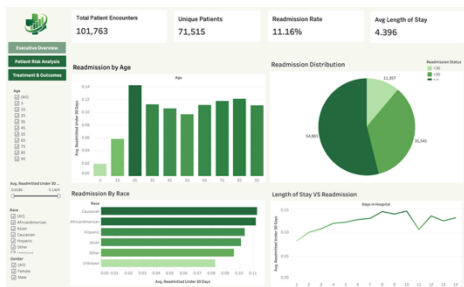
Three-layer decision support suite — built for CMO, CFO & Discharge Planners

1

Executive Overview

What a CEO sees

- KPI header cards: 11.16% readmission rate, 101,763 encounters, Avg. LOS 4.4 days
- Age-segmented readmission bar chart — highlights 75–85 peak and 25–35 secondary peak
- Race breakdown showing Caucasian vs. African-American volume differences
- Global filters: Age Range, Race, Admission Source — updates all views in <1 sec

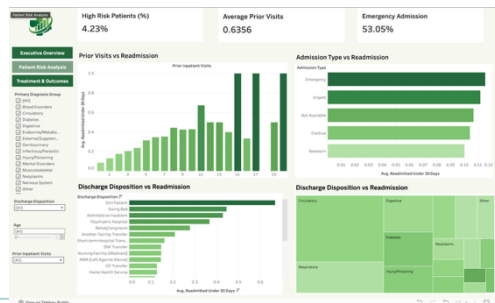


2

Patient Risk Analysis

What an analyst drills into

- Treemap of diagnosis groups — sized by encounter count, colored by readmission rate
- Prior visits bar chart: 0 visits = 10%, 3+ visits = ~50% readmission rate
- Admission type breakdown: Emergency (53%) vs. Urgent vs. Elective
- Interactive: clicking 'Emergency' updates diagnosis treemap for ER-only view



3

Treatment View

Clinical signal deep-dive

- Medication change effect: Changed = 11.82% vs. No Change = 10.59% readmission
- Insulin dosage breakdown: Down 13.9% · Up 13.0% · Steady 11.1% · None 10.0%
- Down-titrated insulin is the highest single treatment risk signal identified
- Filter: Isolate insulin-adjusted patients to view their diagnosis group profile

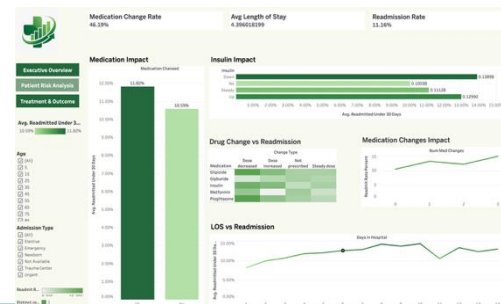


Tableau Dashboard

URL: https://public.tableau.com/app/profile/ayush.mittal4873/viz/DVACapstoneDiabetesReadmissionAnalysis_v2025_3/Dashboard_Executive_Overview?publish=yes

From Insight to Action: Three Targeted Recommendations

Each recommendation traces directly from a statistical finding

1

Targeted Follow-up Program

Insight

Insight (Slide 5): Patients with ≥ 3 prior inpatient visits have a $\sim 50\%$ readmission rate — 4.5 $\times$ the cohort average.

Recommendation

Recommendation: Implement a mandatory 48-hour post-discharge nurse callback program for all patients flagged with ≥ 3 prior inpatient visits. Flag this automatically in the EMR system at admission.

Expected Impact

Expected Impact: High | Feasibility: Easy | Est. prevention of up to 40% of avoidable early returns in this segment

2

Heart-Diabetes Clinical Pathway

Insight

Insight (Slide 5 & 6): Circulatory comorbidities drive $\sim 30\%$ of all 30-day returns, especially in the 75–85 age cohort.

Recommendation

Recommendation: Develop a specialized 'Heart-Diabetes' discharge checklist. Mandate secondary review by a clinical pharmacist for all Circulatory-flagged patients before discharge authorization.

Expected Impact

Expected Impact: High | Feasibility: Medium | Targets the single largest diagnosis driver of readmissions

3

Insulin Stability Signoff Rule

Insight

Insight (Slide 6): Down-titrated insulin = 13.9% readmission rate — the highest single treatment signal. Medication changes broadly elevate risk from 10.59% $\rightarrow$ 11.82%.

Recommendation

Recommendation: Require attending physician signoff if a patient's insulin is down-titrated within 24 hours of planned discharge. Mandate pharmacist review for all 'Changed' medication encounters.

Expected Impact

Expected Impact: Medium | Feasibility: Easy | Directly addresses the highest-risk treatment instability marker

Impact & Value: The Business Case for Action

Quantified savings, efficiency gains, and cost of inaction

\$1.2M

Annual HRRP Penalty
Savings

5% reduction in 30-day readmissions eliminates this much in regulatory penalties per hospital

2,000+

Bed-Days Freed
Per Year

Preventing avoidable returns releases ~0.2 days per patient stay across the full encounter volume

40%

Avoidable Returns
Preventable

Targeted nurse callbacks to the '≥3 prior visits' cohort could prevent up to 40% of early readmissions in that group

Cost of Inaction

Maintaining the current 11.16% readmission rate ensures ongoing HRRP margin compression, declining hospital quality ratings, and continued unnecessary patient harm. Every 30-day readmission that could have been prevented represents a direct failure in the duty of care — and an avoidable financial penalty. The cost of inaction compounds annually.

Limitations & Next Steps

What the data cannot tell us — and what we build next

Current Limitations

No SDOH Data

Dataset lacks ZIP codes, income levels, and Social Determinants of Health — known to be strong hidden drivers of readmission that are absent from our model.

No Mortality Data

Absence of out-of-hospital death records may artificially lower the readmission rate for high-severity patients who never returned because they passed away.

Retrospective Design

10-year historical data (1999–2008) may not reflect current treatment protocols, insurance structures, or post-ACA care patterns at US hospitals.

No Outpatient Follow-up

Dataset only captures inpatient encounters. We cannot measure whether patients received post-discharge care or had access to primary care providers.

Strengthening Findings & Next Steps

Real-Time Predictive Model

- 1 Build an XGBoost classifier that flags high-risk patients at the moment of admission — enabling proactive rather than reactive discharge planning. Incorporate EMR text notes via NLP.

SDOH Data Integration

- 2 Enrich with ZIP-code-level income and healthcare access data to model social risk alongside clinical risk. Map 'Readmission Hotspots' geographically by city and neighborhood.

Extended Dataset Scope

- 3 Integrate post-2008 data to capture ACA-era changes, telemedicine adoption, and updated diabetes treatment protocols. Move from retrospective to near-real-time analysis.

Live Tableau + EMR Integration

- 4 Connect Tableau to live EMR feeds for real-time dashboard updates. Move from static historical reporting to proactive prevention — alert discharge planners at the bedside.